

SECTION 5.4

EIGENVECTORS AND LINEAR TRANSFORMATIONS

ELEMENTARY LINEAR ALGEBRA

(MATH 2250-03)

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The goal of this section is to understand the matrix factorization $A = PDP^{-1}$ as a statement about linear transformations. We shall see that the transformation $x \mapsto Ax$ is essentially the same as the very simple mapping $u \mapsto Du$, when viewed from the proper perspective. A similar interpretation will apply to A and D even when D is not a diagonal matrix.

Recall from Section 1.9 that any linear transformation T from $\mathbb{R}^n$ to $\mathbb{R}^m$ can be implemented via left-multiplication by a matrix A , called the *standard matrix* of T . Now, we need the same sort of representation for any linear transformation between two finite-dimensional vector spaces.

THE MATRIX OF A LINEAR TRANSFORMATION

Let V be an n -dimensional vector space, let W be an m -dimensional vector space, and let $T : V \rightarrow W$ be any linear transformation. To associate a matrix with T , choose two bases, $\mathcal{B}$ and $\mathcal{D}$, for V and W , respectively. Then, given any $x \in V$, the coordinate vector $[x]_{\mathcal{B}}$ is in $\mathbb{R}^n$ and the coordinate vector of its image, $[T(x)]_{\mathcal{D}}$, is in $\mathbb{R}^m$. The connection between $[x]_{\mathcal{B}}$ and $[T(x)]_{\mathcal{D}}$ is easy to find. Namely, write $\mathcal{B} = \{b_1, \dots, b_n\}$ so that

$$\begin{aligned} x &= c_1 b_1 + \dots + c_n b_n \\ &= \begin{bmatrix} b_1 & \dots & b_n \end{bmatrix} [x]_{\mathcal{B}}. \end{aligned}$$

Then, by the linearity of T , we have

$$\begin{aligned} T(x) &= T(c_1 b_1 + \dots + c_n b_n) \\ &= c_1 T(b_1) + \dots + c_n T(b_n). \end{aligned}$$

So, it follows from the linearity of the coordinate mapping $y \mapsto [y]_{\mathcal{D}}$ that

$$[T(x)]_{\mathcal{D}} = c_1 [T(b_1)]_{\mathcal{D}} + \dots + c_n [T(b_n)]_{\mathcal{D}}.$$

So, putting it all together, we reach $[T(x)]_{\mathcal{D}} = M[x]_{\mathcal{B}}$ where M is the matrix

$$M = \begin{bmatrix} [T(b_1)]_{\mathcal{D}} & \dots & [T(b_n)]_{\mathcal{D}} \end{bmatrix}.$$

This matrix M is a matrix representation of T called *the matrix for T relative to the bases $\mathcal{B}$ and $\mathcal{D}$* .

In the common case where W is the same as V and the basis $\mathcal{D}$ is the same as $\mathcal{B}$, the matrix M above is called the *matrix for T relative to $\mathcal{B}$* , or simply the $\mathcal{B}$ -matrix for T , and is denoted by $[T]_{\mathcal{B}}$. The $\mathcal{B}$ -matrix for $T : V \rightarrow V$ satisfies

$$[T(x)]_{\mathcal{B}} = [T]_{\mathcal{B}}[x]_{\mathcal{B}} \quad \text{for all } x \in V.$$

Example 1. Suppose $\mathcal{B} = \{b_1, b_2\}$ is a basis for a vector space V and $\mathcal{D} = d_1, d_2, d_3$ is a basis for a vector space W . Let $T : V \rightarrow W$ be a linear transformation with the property that

$$T(b_1) = 3d_1 - 2d_2 + 5d_3 \quad \text{and} \quad T(b_2) = 4d_1 + 7d_2 - d_3.$$

Find the matrix M for T relative to $\mathcal{B}$ and $\mathcal{C}$.

To solve this example, notice that the $\mathcal{D}$ -coordinate vectors of the images of b_1 and b_2 under T are $[T(b_1)]_{\mathcal{D}} = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix}$ and $[T(b_2)]_{\mathcal{D}} = \begin{bmatrix} 4 \\ 7 \\ -1 \end{bmatrix}$. Hence, our answer is simply $M = \begin{bmatrix} 3 & 4 \\ -2 & 7 \\ 5 & -1 \end{bmatrix}$.

LINEAR TRANSFORMATIONS ON $\mathbb{R}^n$

In an applied problem involving $\mathbb{R}^n$, a linear transformation T usually appears first as a matrix transformation, $x \mapsto Ax$. If A is diagonalizable, then there is a basis $\mathcal{B}$ for $\mathbb{R}^n$ consisting of eigenvectors of A . The theorem below shows that, in this case, the $\mathcal{B}$ -matrix for T is diagonal. So, diagonalizing A amounts to finding a diagonal matrix representation of $x \mapsto Ax$.

Theorem 1 (Diagonal Matrix Representation). *Suppose $A = PDP^{-1}$, where D is a diagonal $n \times n$ matrix and P is an invertible $n \times n$ matrix. If $\mathcal{B}$ is the basis for $\mathbb{R}^n$ formed from the columns of P , then D is the $\mathcal{B}$ -matrix for the transformation $x \mapsto Ax$.*

Example 2. Define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(x) = Ax$, where $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$. The matrix A satisfies $A = PDP^{-1}$, where

$$P = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}.$$

Find a basis $\mathcal{B}$ for $\mathbb{R}^2$ with the property that the $\mathcal{B}$ -matrix for T is a diagonal matrix.

To solve this example, notice that the columns of P —call them b_1 and b_2 , respectively—are eigenvectors of A . By the above theorem, D is the $\mathcal{B}$ -matrix for T , where $\mathcal{B} = \{b_1, b_2\}$. The mappings $x \mapsto Ax$ and $[x]_{\mathcal{B}}D[x]_{\mathcal{B}}$ describe the same linear transformation, relative to different bases.

SIMILARITY OF MATRIX REPRESENTATIONS

If $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a linear transformation defined by $T(x) = Ax$, and if $\mathcal{B}$ is any basis for $\mathbb{R}^n$, then the $\mathcal{B}$ -matrix for T is similar to A . In fact, if P is the matrix whose columns come from the vectors in $\mathcal{B}$, then as

$$\begin{aligned} [T]_{\mathcal{B}} &= \begin{bmatrix} [T(b_1)]_{\mathcal{B}} & \dots & [T(b_n)]_{\mathcal{B}} \end{bmatrix} \\ &= \begin{bmatrix} [Ab_1]_{\mathcal{B}} & \dots & [Ab_n]_{\mathcal{B}} \end{bmatrix} \\ &= \begin{bmatrix} P^{-1}Ab_1 & \dots & P^{-1}Ab_n \end{bmatrix} \\ &= P^{-1}A \begin{bmatrix} b_1 & \dots & b_n \end{bmatrix} \\ &= P^{-1}AP, \end{aligned}$$

we may conclude that $[T]_{\mathcal{B}} = P^{-1}AP$. Thus, the set of all matrices similar to a matrix A coincides with the set of all matrix representations of the transformation $x \mapsto Ax$.